

LINGUAL SENSE

Deep Learning for Language Detection Across Texts



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PROJECT OVERVIEW

OBJECTIVE:

To build a model that can automatically identify the language of a given text. Language identification is essential for various applications, including machine translation, multilingual document tracking, and electronic devices (e.g., mobiles, laptops).

OUTCOMES:

- A trained deep learning model capable of predicting the language of input text
- Improved accuracy in language detection for multilingual documents.



DATASET OVERVIEW AND KEY INSIGHTS

Structure: 2 columns: Text, Language.

Size: 32,337 rows, 2 columns.

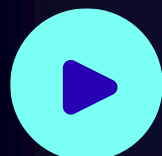
Strengths: Comprehensive textual data with associated language labels.

REASON OF CHOOSING THE DATASET:

Comprehensive Textual Data: The dataset contains 32,337 rows of detailed text entries labeled with their respective languages, offering a rich source for textual and linguistic analysis.

Language Identification: Includes a 'Language' column that enables classification and segmentation tasks, which are essential for natural language processing and supervised model training.

Scalable for NLP Tasks: The large dataset size ensures robust training data, making it suitable for tasks like sentiment analysis, translation models, and text summarization.



METHODOLOGY

- **DATA PREPROCESSING**

Preparing raw data by cleaning, formatting, and organizing it for analysis.

- **FEATURE EXTRACTION**

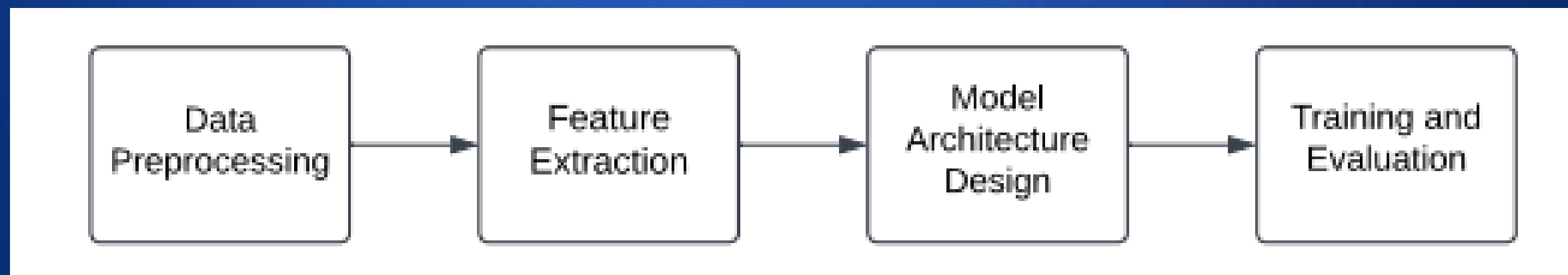
Converting data into meaningful representations suitable for modeling.

- **MODEL ARCHITECTURE EXTRACTION**

Designing and implementing algorithms or architectures to solve the problem.

- **TRAINING AND EVALUATION**

Training the model on data and measuring its performance using relevant metrics.



EXPLORATORY DATA ANALYSIS

	Text	Language
count	32337	32337
unique	32117	30
top	haec commentatio automatica praeparata res ast...	English
freq	48	2385

Inspected the structure and quality of the data using `df.info()` and `df.describe()`

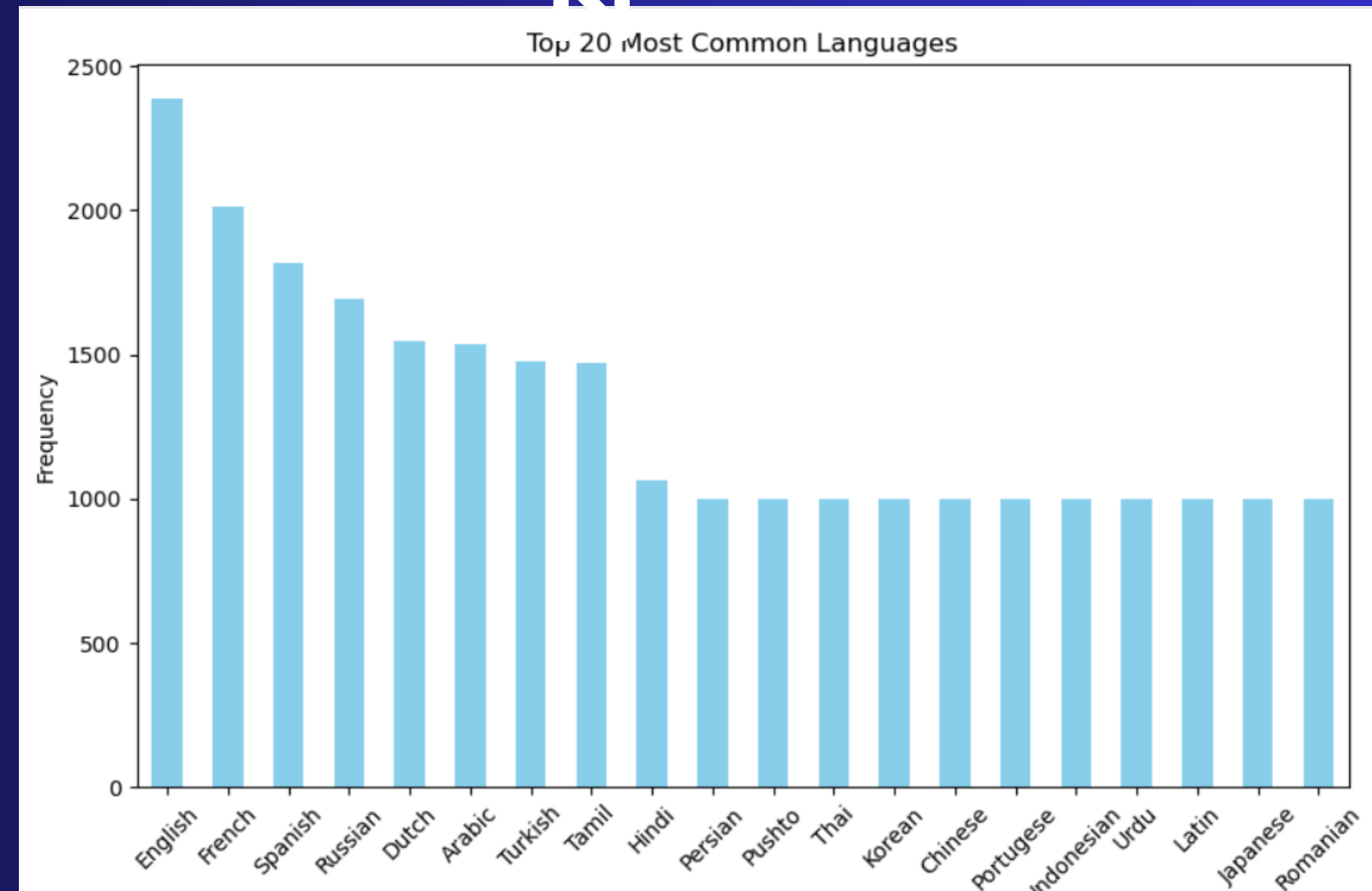
KEY FINDINGS:

- out of 32337 rows there are 32117 unique rows
- There are 30 unique languages
- English dominating other languages with 2385 entries



VISUALIZATION

Language	
English	2385
French	2014
Spanish	1819
Russian	1692
Dutch	1546
Arabic	1536
Turkish	1474
Tamil	1469
Hindi	1063
Persian	1000
Pushto	1000
Thai	1000
Korean	1000
Chinese	1000
Portugese	1000
Indonesian	1000
Urdu	1000
Latin	1000
Japanese	1000
Romanian	1000
Swedish	1000
Estonian	1000
Portugeese	739
Italian	698
Sweedish	676
Malayalam	594
German	470
Danish	428
Kannada	369
Greek	365



- English is the most frequent language , making up a significant portion (7.38%) of the total entries.
- Greek is the least frequent language in the dataset, with only 365 entries, constitutes just 1.13% of the total dataset.

DATA PREPROCESSING

1.REMOVED DUPLICATES:

Duplicates can artificially inflate the representation of certain data points, leading to model bias. Hence,Identical text entries were identified and removed.

2.CLEANING THE TEXT DATA:

Regex-based cleaning was then applied to standardize the text by removing punctuation, non-alphanumeric characters, and extra spaces, resulting in a more uniform structure.

3.NORMALIZATION

Finally, normalization was performed by converting all text to lowercase, ensuring consistency and reducing unnecessary variability in the dataset

FEATURE EXTRACTION

1.TOKENIZATION:

The text data was transformed into numerical sequences, where each unique word was assigned an integer. This converts raw text into numerical form, enabling the model to process it.

2.PADDING:

To ensure all sequences had the same length, padding was applied to extend or truncate sequences to a maximum length of 100 words.

Padding ensures consistent input dimensions, which is essential for batch processing in neural networks.

3.LABEL

ENCODING:

Language labels were converted into numerical form using LabelEncoder from sklearn. Each language was assigned a unique integer, making the categorical labels compatible with the model.

MODEL ARCHITECTURE

- The architecture is a sequential neural network built using Gated Recurrent Units (GRUs), which are efficient for handling sequential data like text. The focus is on extracting meaningful patterns from sequences to predict the language accurately.

LAYERS OF

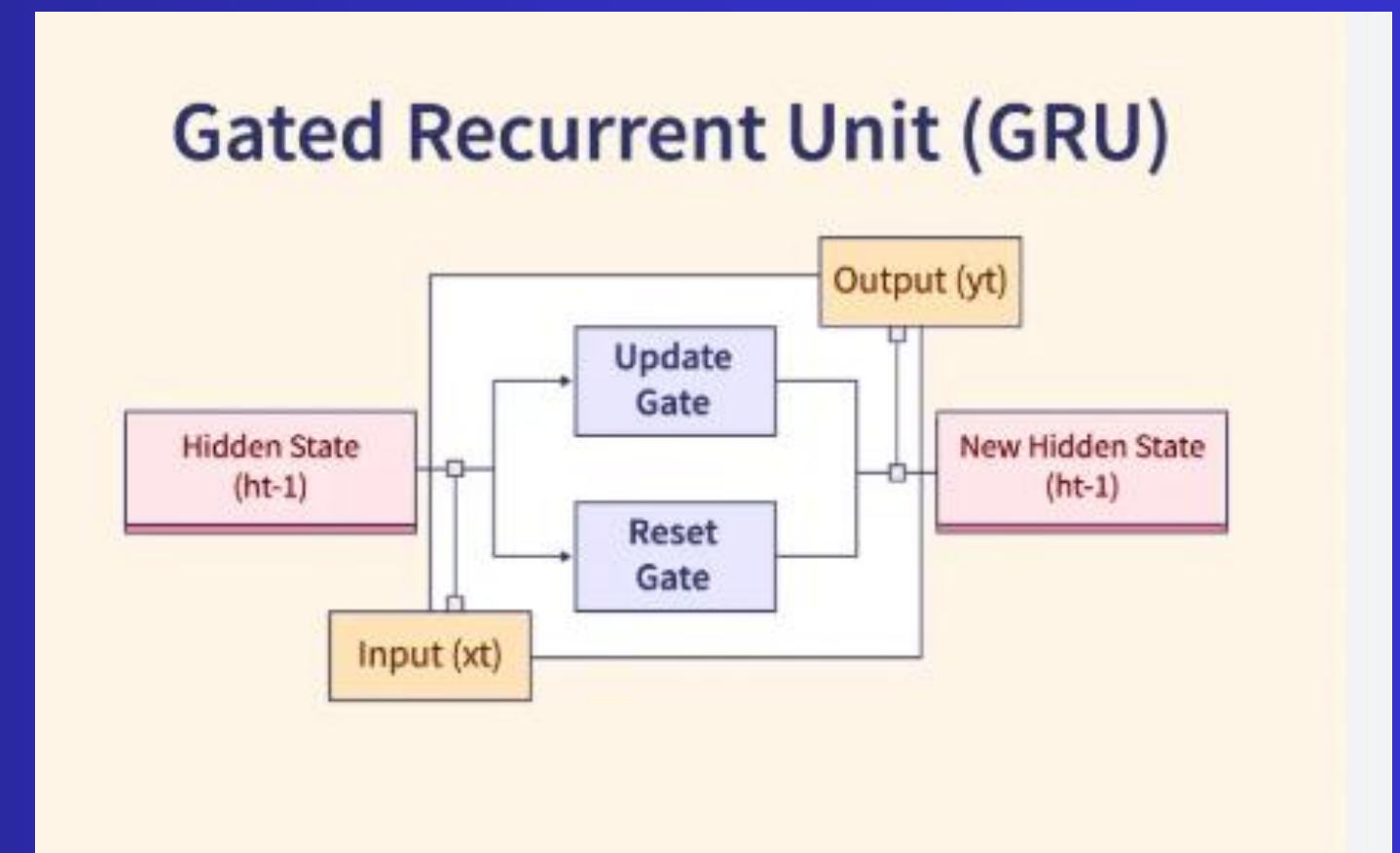
- **GRU:**
INPUT

- **EMBEDDING**

LAYER

LAYERS AND DROPOUT

- **OUTPUT LAYERS**
LAYER



TRAINING AND EVALUATION

During training, the dataset was divided into 80% for training and 20% for testing, with 20% of the training data used for validation. This approach ensures that the model's generalization ability is fairly evaluated.

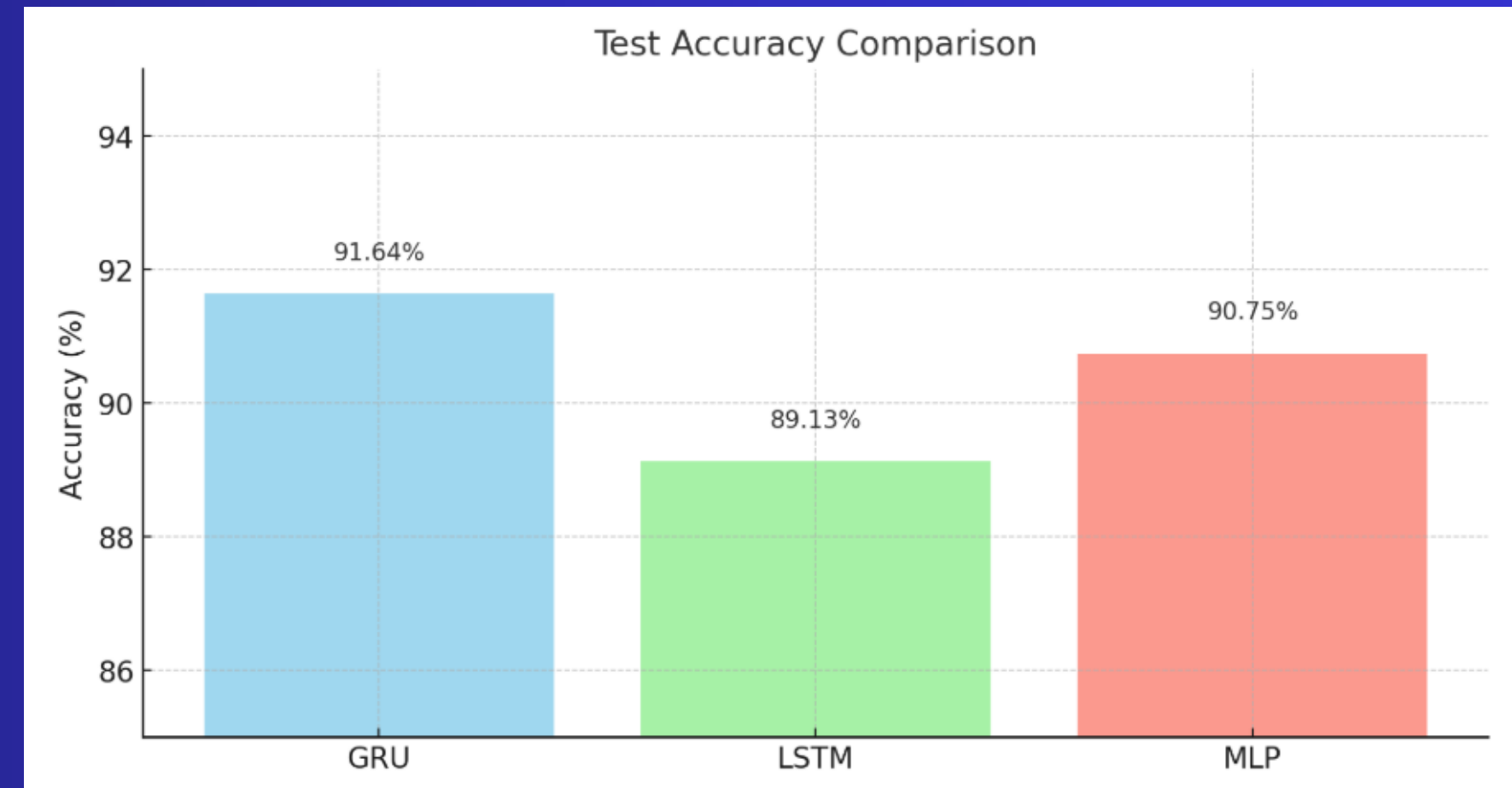
We trained three models: GRU, LSTM, and MLP, to evaluate their performance. GRU outperformed the others with the highest accuracy and efficiency, making it the best fit for our task.

While LSTM also performed well, it was more resource-intensive. MLP, being less suited for sequential data, had the lowest accuracy. Ultimately, GRU was chosen for its balance of performance and speed.

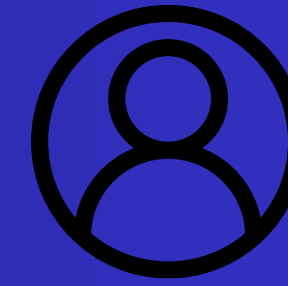
Model	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss	Test Accuracy	Test Loss
GRU	89.24%	0.3120	91.89%	0.4303	91.64%	0.4089
LSTM	90.76%	0.3245	89.53%	0.4085	89.13%	0.4239
MLP	89.23%	0.3360	90.83%	0.3661	90.75%	0.3447

RESULTS

- **GRU:** Best overall accuracy and balanced performance, making it ideal for the task.
- **LSTM:** Slightly lower accuracy but still a strong performer, particularly for more complex sequences.
- **MLP:** Competitive accuracy and the lowest test loss, indicating robustness for simpler tasks.



USER INTERFACE



Lingual Sense (Language Detection App)

Detect the language of a paragraph using a GRU-based deep learning model!

Input a Paragraph:

Enter your paragraph here...

Predict Language

Lingual Sense (Language Detection App)

Detect the language of a paragraph using a GRU-based deep learning model!

Input a Paragraph:

இன்று உலகம் முழுவதும் தொழில்நுட்பம் முக்கியமான பங்கு வகிக்கின்றது. இணையதளம், செயற்கை நுண்ணறிவு மற்றும் தகவல் தொடர்பு தொழில்நுட்பங்கள் முன்னேறியுள்ளதால், நாம் பல விஷயங்களை எளிதாகச் செய்ய முடிகின்றது. ஆனால், இதன் பின்புலத்தில் உள்ள சவால்களை அணுகும் போது, நம் சமூகத்தில் வருகிற மாற்றங்களை நாம் நன்கு புரிந்து

Predict Language

Detected Language: Tamil

CHALLENGES

1.DATA

IMBALANCE:

Certain languages had significantly more data than others, leading to potential bias in model predictions.

2.OVERFITTING

• Preventing overfitting, especially with a large number of languages and limited data for some classes, required careful regularization.

3.MULTILINGUAL NOISE IN

DATASET:

Cleaning mixed-language text entries posed challenges for consistent language labeling

FUTURE SCOPE

- Expanding the dataset to include more underrepresented and rare languages to improve model versatility.
- Deploying the model as a real-time application for detecting text language in social media, web content, and more.
- Leveraging knowledge from high-resource languages to improve predictions for low-resource ones.



CONCLUSION



- **Successfully developed a deep learning model for accurate language detection across texts**
- **Achieved robust performance with high accuracy and reliable evaluation metrics**
- **Demonstrated the importance of language detection in multilingual document processing and its applications in real world scenarios**

THANK YOU

